

Annual Network Prices Report

1 July 2005 – 30 June 2006

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1. Introduction

1.1 Background

Country Energy has compiled this report to outline how network prices are set and what path these prices are expected to take for the forthcoming year.

The economic regulatory arrangements in New South Wales governed by the Independent Pricing and Regulatory Tribunal (IPART) require Country Energy to publish an Annual Network Prices Report which establishes a process of price notification and review by IPART for annual price changes.

This report complies with the requirements of the Determination, under Annexure 17 Annual Prices Document Information Requirements, and specifically addresses the following:

- ✓ Prices for prescribed distribution services
- ✓ Forthcoming changes in network prices
- ✓ Compliance with the regulatory arrangements relating to limits on price and revenue movements
- ✓ Impacts of the proposed changes on customers
- ✓ Pricing principles and the network strategy statement
- ✓ Expected levels of service and projected capital expenditure
- ✓ Expected consumption for major customer classes and maximum demand for the coming year.

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1.2 Our Proposal

Country Energy is proposing an overall increase in average prices for prescribed distribution services from 1 July 2005 of 4.8 per cent nominal. We are also proposing to simplify and reduce the number of network prices by rationalising nine network prices.

Country Energy is not proposing any changes to existing price structures, application of these price structures or associated criteria. The number of prices applicable to new connections, and the number of prices classed as obsolete and not available to new connections will not change.

1.3 Our Business

From 1 July 2005, Country Energy merged its operations with Australian Inland. Country Energy already operated Australia's largest power network, with 185,000 kilometres of powerlines serving three quarters of New South Wales – from Bega to Tweed Heads, west to Bourke, Cobar, and Hay, and south to the River Murray.

Australian Inland's 9,000 kilometres of powerlines span almost 20 per cent of the State, serving Broken Hill, Tibooburra, Wilcannia, White Cliffs, Wentworth, Silverton, Balranald, Menindee and Moulamein.

The two organisations combined have almost 200,000 kilometres of powerlines serving approximately 95 per cent of New South Wales' land mass.

Country Energy is now one of Australia's leading utility corporations, with around 3,300 employees and 780,000 customers. Country Energy has a decentralised regional structure with no single head office, ensuring that decisions are made locally and wages are spent where they are earned.

The figure below illustrates our distribution area, which covers approximately 95 per cent of New South Wales.



Figure 1 Country Energy's distribution area

2 Prices for Prescribed Distribution Services

Country Energy provides prescribed distribution network services to all electricity customers connected within our distribution area. Customers may choose any retailer.

Prescribed distribution network services are those services performed by Country Energy to provide access to our power supply network. Those services are principally the ownership, planning, design, construction, operation and maintenance of the

electricity distribution network - consisting of equipment such as poles, wires and substations - to supply a safe and reliable power supply.

The conditions under which these network services are provided are contained in Country Energy's Standard Form Customer Connection Contract. A copy of this agreement is available at www.countryenergy.com.au or on request by phoning 13 23 56.

Network prices are a use of system charge for the use of Country Energy's electricity distribution network and also include a charge for the use of the transmission system.

Network prices are subject to annual review in accordance with pricing determinations by IPART. The Network Price List is available on Country Energy's website www.countryenergy.com.au or on request. The network price list contains network prices, monopoly fees and miscellaneous charges effective from 1 July 2005.

Network access is available under conditions contained in our Standard Form Customer Connection Contract. A negotiated network connection agreement is also available where specific contractual conditions are required.

Prescribed distribution services supplied by Country Energy and covered by this Network Prices Report include:

- Distribution use of system services
- Private powerline inspections and customer installation inspections
- Miscellaneous services
- Monopoly services associated with contestable works
- Emergency recoverable works.

3 Forthcoming changes in network tariffs

Country Energy is proposing an average increase in prices for prescribed distribution services of 4.8 per cent. We believe our proposal will assist in establishing a financially sustainable level of network revenue that will ensure Country Energy has sufficient resources to meet its obligations, maintain its significant asset investment and to continue to efficiently deliver prescribed distribution services to meet increasing customer expectations.

Country Energy is not proposing any changes to:

- ✓ existing price structures
- ✓ application of these price structure
- ✓ associated criteria
- ✓ the number of prices applicable to new connections
- ✓ the number of prices classed as obsolete and not available to new connections.

Country Energy is proposing that the unification of nine network prices would further reduce the number of network prices, and also reduce the number of obsolete network prices. The following figures demonstrate the rationalisation of prices and network price codes. The current rates for each of the network prices together with the proposed change are tabled below.

The unification of prices has been completed within limits on network price movements as defined in the determination. Below is a summary of the proposed

rationalisation of tariffs and the average increases required to unify each group. For full details of the tariffs and components please refer to Country Energy's network price list available from www.countryenergy.com.au or on request.

Figure 2 Proposed rationalisation of network prices

Obsolete tariff	Rationalised to equal non-obsolete tariff	Average increase of obsolete tariff	Average increase of non-obsolete tariff
BLNT3SM	BLNT3AU	4.00%	4.00%
BLNC5CR	BLNC5AR	4.10%	4.00%
BLND1NO	BLND1CO	-19.40%	4.00%
BLNN4CR	BLNN2AR	-23.00%	4.00%
BLNC2CR	BLNC2AR	4.00%	4.00%
BLNN2SQ	BLNN2AR	6.40%	4.00%
BLNN2NQ	BLNN2AR	4.00%	4.00%
BLNN2CR	BLNN2AR	0.50%	4.00%
BLNT3SM	BLNT3AR	4.00%	4.00%

Country Energy has prepared this proposal in accordance with the determination. All changes proposed are within the limits on price movements set out in the determination and are discussed further throughout this document.

4 Regulatory compliance

4.1 Overview

IPART has determined the regulatory arrangements that apply to prescribed distribution services over the regulatory period and how it will regulate charges for prescribed distribution services and other fees that electricity distributors are entitled to charge over the regulatory period. This annual network prices report explains how Country Energy's prices meet the regulatory arrangements, including limits on network price movements, weighted average price cap control formula and provide for the recovery of transmission related payments. Further detail on each of these items is contained below.

4.1.1 Weighted average price cap

Charges for prescribed distribution services (DUOS prices, miscellaneous charges, monopoly fees and charges for recoverable works for emergency services) are regulated in accordance with a weighted average price cap (WAPC) formula which includes a CPI-X factor. The WAPC operates by restricting the (weighted) average change in Country Energy's prices to a limit (the 'X-factor') determined by IPART. The CPI-X factor is used to achieve a price path that ensures revenue raised reflects underlying costs.

The average allowable increases in distribution prices (or X-factor) has been limited by IPART to 2.5 per cent real in 2005/06. The x-factor is contained in the formula for the WAPC and determines how prices for prescribed distribution services can change in real terms over the regulatory period.

In setting prices for the upcoming year, Country Energy must ensure that the average price change relative to the prices it is charging satisfies the constraint. Country Energy has applied the weighted average price control formula set out in clause 5.2

of the determination to each of the DUOS prices, miscellaneous service charges and monopoly service charges for 2004/05.

As part of the annual price setting process, IPART was provided with all supporting calculations and information including audited quantities necessary to demonstrate compliance with all of the requirements of clause 5 of the determination. The application of the weighted average price cap formula results in an average price increase of 4.8 per cent for prescribed distribution services in accordance with the determination, and IPART has verified compliance with the determination.

4.1.2 Transmission-related cost recovery arrangements

IPART allows Country Energy to recover transmission-related costs by setting transmission cost recovery (TUOS) prices to recover -

- Transmission charges paid to transmission network service providers
- Avoided TUOS payments to embedded generators calculated in accordance with the Code
- Inter-distributor transfer payments to other distributors.

Clause 6 of the determination requires Country Energy to demonstrate to IPART compliance with transmission cost recovery requirements. As part of the 2005/06 price approval process, IPART has been provided with the expected cost of transmission related payments. In accordance with clause 3.4 of the determination, Country Energy develops transmission cost recovery prices, using the methodology outlined in Country Energy's Network Strategy Statement, to recover the expected cost of transmission-related payments.

4.1.3 Limits on network price movements

Each individual network price is subject to price limits. For the period commencing 1 July 2005, each individual tariff can not increase by more than CPI+4.5 per cent (or 6.8 per cent). In addition to this constraint, the fixed component of a residential network tariff cannot increase by more than \$30 per annum.

As part of the annual price setting process, IPART was provided with all supporting calculations and information necessary to demonstrate compliance with all of the requirements of clause 7 of the determination. Country Energy's residential and non-residential network price increases for 2005/06 do not exceed 6.8 per cent when compared to the equivalent network price for 2004/05 and the fixed charge of any network price for residential customers has not increased by more than \$30. IPART verifies that the prices comply with the determination.

5 Impact of forthcoming changes to network tariffs

This section demonstrates the impact of the forthcoming changes in Network Tariffs on typical customers' bills, including disclosing forecast average prices (based on typical account categories).

The following table demonstrates the average impact of the proposed prices to the residential and business customer classes. Displayed below are the average increases expected for each of the consumption types for residential and business network prices. These include standard supply, time of use, controlled load, and demand network prices for business customers.

Figure 3 Average increases for domestic and business customers

		Average annual MWh	Average annual current account	Average increase per customer	Average increase (%)	Average current c/KWh	Average proposed c/KWh
Residential	Standard	5.19	\$379.10	\$23.17	6.11%	7.30	7.75
	Time of use	15.63	\$795.47	\$50.21	6.31%	5.09	5.41
	Controlled load	3.14	\$50.97	\$2.26	4.44%	1.62	1.69
	Total	4.36	\$242.66	\$14.48	5.97%	5.57	5.90
Non-Residential	Standard	15.90	\$1,480.72	\$61.48	4.15%	9.31	9.70
	Time of use	21.76	\$1,292.65	\$55.60	4.30%	5.94	6.20
	Controlled load	5.02	\$899.92	\$35.78	3.98%	17.91	18.62
	Demand	1,178.17	\$47,882.31	\$1,770.02	3.70%	4.06	4.21
	Total	48.36	\$2,649.33	\$104.28	3.94%	5.48	5.69

The average domestic customer in Country Energy's distribution area will see an increase of approximately \$14 or 5.97 per cent for the 2005/06 year, based on an annual consumption of 4.4 MWh.

The average business customer in Country Energy's distribution area will see an increase of approximately \$104 or 3.94 per cent for the 2005/06 year, based on an annual consumption of 48.4 MWh.

A typical urban residential customer living in Country Energy's distribution area would generally be connected to the following network prices:

- BLNN2AU: Urban residential non-TOU
- BLNC1AU: Urban residential controlled load 1

Provided below is an analysis of the impacts of price increases for a low usage customer and a typical usage customer.

Figure 4 Impact of price increases for typical urban residential customers of Country Energy

Customer Type	% Controlled load	2004/05 Quarterly network bill	2005/06 Quarterly network bill	Increase in quarterly network bill
Low usage (3,500 KWh)	40%	\$63	\$67	\$4
Typical usage (7,500 KWh)	40%	\$103	\$110	\$7

A typical rural residential customer living in Country Energy's distribution area would generally be connected to the following network prices:

- BLNN2AR: Rural residential non-TOU

➤ BLNC1AR: Rural residential controlled load 1

Provided below is an analysis of the impacts of price increases for a low usage customer and a typical usage customer.

Figure 5 Impact of price increases for typical rural residential customers of Country Energy

Customer Type	% Controlled load	2004/05 Quarterly network bill	2005/06 Quarterly network bill	Increase in quarterly network bill
Low usage (3,500 KWh)	40%	\$76	\$79	\$3
Typical usage (7,500 KWh)	40%	\$125	\$130	\$5

A typical small business customer operating in Country Energy's distribution area would generally be connected to the following network price:

➤ BLNN1AU: Urban business non-TOU

Provided below is an analysis of the impacts of price increases for a customer that consumes 20 MWh per annum and a customer that consumes 40 MWh per annum.

Figure 6 Impact of price increases for typical business customers of Country Energy

Customer Type	2004/05 Monthly network bill	2005/06 Monthly network bill	Increase in monthly network bill
20 MWh per annum	\$147	\$153	\$6
40 MWh per annum	\$282	\$293	\$11

6 Pricing principles and Network Strategy Statement

Clause 12.2 of the *NSW Electricity Distribution Pricing 2004/05 to 2008/09 Determination* (the determination) requires that in setting prices for prescribed distribution services for each year of the regulatory control period, Country Energy must comply with the Pricing Principles as set out in Annexure 13.

Annexure 13 of the determination sets out the pricing principles that Country Energy must consider when setting prices. This section summarises how these principles have been integrated into Country Energy's pricing methodology.

The Network Strategy Statement provides details of the level and structure of current prices for prescribed distribution services, the current standards of service provided, the methodology used to derive prices and their current cost base, and the medium term directions for prices and standards of service. The approach used by Country Energy to set current prices is equally applicable to the 2005/06 prices. Our Network Strategy Statement is available at www.countryenergy.com.au or on request.

Country Energy confirms that it has applied the Pricing Principles as outlined below to the network prices, and the prices are also consistent with the information contained within our Network Strategy statement.

1. Prices are to be consistent with the form of regulation, including any price limits on network tariffs determined by the Tribunal.

This Annual Prices Report demonstrates compliance with the regulatory requirements for prices submitted to IPART for verification and approval. Country Energy believes it has fully complied with the requirements of the Determination as demonstrated in the information provided to IPART in the annual pricing approval information package. In summary: ✓ The weighted average price change in DUOS tariffs is within the allowed weighted average change of 4.8 per cent ✓ Average increases in network tariffs are below the allowed change of 6.8 per cent and all increases in the fixed component of residential tariffs do not exceed \$30 Country Energy has submitted to IPART estimates in relation to components of transmission related payments in the annual pricing approval information package.	References: Section 1.7 – Country Energy's Network Strategy Statement Clause 3 – The Determination
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2. Prices should be based on a well-defined and clearly explained methodology.

The Network Strategy Statement provides details on the methodology used by Country Energy to develop prices. An overview of this pricing methodology to set network prices is provided below. Country Energy has separated each network price into a DUOS price for providing distribution use of system services to distribution customers and a separate transmission cost recovery (TUOS) price. The methodology adopted by Country Energy for setting network prices is summarised as follows - 1. Project total network revenue requirements, taking into account distribution revenue derived from the WAPC form of regulation, the cost of transmission related payments, transmission under and over recovery balance and the forecast growth in the network 2. Model fully distributed costs 3. Model the marginal and stand-alone cost of supply 4. Assess the existing network prices against the cost of supply 5. Establish medium-term price paths to move existing network prices steadily toward the subsidy free zone, whilst complying with the economic regulatory arrangements as set by IPART in the determination, and Country Energy's medium term network pricing strategy to progressively equalise prices across the network.	References: Section 2 – Country Energy's Network Strategy Statement
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3. Price development should incorporate an analysis of the cost of service provision that includes:

(a) definition of the classes of service provided and the parameters by which the quantum and standard of service in each class are measured

i. Definition of the classes of service provided

Distribution services are provided to a range of customer classes. Customer classes have been established taking into consideration: ✓ The legacy of previous pricing structures ✓ Existing metering and associated data ✓ Cost effectiveness of metering options ✓ Connected voltage level. The current definitions of network customer classes can be found in Country Energy's Network Price List, as amended from time to time. This list is updated each year or as required in consultation with IPART. The definition of the classes of prescribed distribution service provided by Country Energy is outlined further in the Network Strategy Statement.	References: Country Energy's Network Price List Section 2.3 – Country Energy's Network Strategy Statement
ii. Definition of the parameters by which the quantum and standard of service in each class are measured	
To monitor Country Energy's performance in meeting customer needs and its corporate strategies, reliability performance indicators are maintained as listed below: ✓ SAIDI - Represents the minutes each customer is without supply for the year when averaged over all customers on the network or part of the network where this is specified ✓ SAIFI - Represents the number of supply interruptions each customer experiences for the year when averaged over all customers on the network or part of the network where this is specified ✓ CAIDI - Represents the average duration of a supply interruption in minutes, for the whole network or part of the network where this is specified. Standards are appropriate as a guide for corporate behaviour and determine the levels of capital and operating expenditure required. They are therefore closely linked to pricing for prescribed distribution services. Section 7.2 of this annual prices report provides the 2005/06 SAIDI objectives for Country Energy.	References: Section 4.3 – Country Energy's Network Strategy Statement
(b) an examination of the cost elements that arise from the use, operation and expansion of the network	
The components of Country Energy costs that arise from the provision of prescribed distribution services include, operating and maintenance costs, return of capital (depreciation), return on capital, regulatory asset base and valuation and transmission-related costs. These cost components are further detailed in the Network Strategy Statement	References: Section 2.6 – Country Energy's Network Strategy Statement
(c) for each class of service and each cost element, identification of the relationship between the quantum and standard of service provided and the level of current and future cost	
The level of the cost of providing prescribed distribution services to network customers and the projected costs to 2009 for operating and maintenance costs, depreciation (return of capital) and return on capital is provided in the determination. Country Energy has broken down its reliability objectives into urban, short rural and long rural feeders, which broadly corresponds to the standard customer classes. The relationship between cost and service levels is outlined in the Network Strategy Statement.	References: Sections 2.3.3, 2.6, 4.3– Country Energy's Network Strategy Statement
(d) an allocation of existing and future network costs to service classes	

Country Energy has adopted the fully distributed cost approach to allocate all network costs to each network customer class and large network customers based on the assets required to serve the customers and their usage of distribution system services. A detailed description of the approach is provided in the Network Strategy Statement.	References: Section 2– Country Energy’s Network Strategy Statement
(e) the translation of allocated costs into service prices at the defined service standard,	
The key principle objectives used by Country Energy in setting network prices are revenue recovery, economic signals, simplicity, stability and equity. We endeavour to satisfy all the above objectives in our approach to setting network prices. However, it is well recognised that these objectives may conflict with each other to some extent and the development of pricing principles requires a balance to be struck between them. There is an important role for judgement in translating the results of the cost modelling into prices. The translation of allocated costs into prices and the principles adopted is described in the Network Strategy Statement.	References: Section 2– Country Energy’s Network Strategy Statement
(f) estimates of the range of subsidy-free prices for each service class	
Country Energy has developed a marginal and standalone cost of supply model for the purpose of estimating the range of subsidy free prices. The Network Strategy Statement provides details of the methodology used and the incremental cost and the standalone cost of supplying distribution services to customers connected to the Country Energy network. Appendix B of the Network Strategy Statement demonstrates the relationship between the current network prices and marginal and stand-alone costs over a range of consumption in each customer class.	References: Sections 2.2, 2.10. Appendix B– Country Energy’s Network Strategy Statement
4. Prices are to signal the economic costs of service provision, by:	
(a) being subsidy free (greater than incremental costs and less than stand-alone costs)	
The graphs in Appendix B of the Network Strategy Statement show that network prices generally lie above the marginal cost of supply and below the lowest cost standalone alternative throughout the range of consumption. For some customer classes, Country Energy’s network prices fall below the marginal cost of supply at low consumption levels. The range of subsidy free prices for each customer class and the methodology applied is described in the Network Strategy statement.	References: Sections 2.2, 2.10, Appendix B– Country Energy’s Network Strategy Statement
(b) having regard to the level of available service capacity	
Country Energy has not developed congestion prices for 2005/06.	
(c) signalling the impact of additional usage on future investment costs.	
The long run marginal cost (LRMC) approach to pricing achieves economic efficiency by signalling the future cost of the next increment of network asset investment. Country Energy strives to send customers accurate network price signals, which reflect the marginal costs of supply. However the LRMC approach to pricing is difficult to implement and pricing based on economically efficient marginal costs does not recover long run average costs, except where there is network congestion. This creates significant tension between economically efficient prices and prices necessary for commercial sustainability. For further information, please refer to the Network Strategy Statement.	References: Sections 2.3.2, 2.7, 2.8, 2.11– Country Energy’s Network Strategy Statement

5. Where prices based on 'efficient' incremental costs under-recover allowed revenues, the shortfall should be made up in a manner that minimises the effect on consumption and investment while having regard to the impact on users

A practical approach to minimising distortions is to recover the gap between marginal and average costs in a manner which does not vary between locations, contains a fixed component and to the extent that a variable component is necessary, includes both energy and demand components. The prices of the great majority of customers are averaged in a conventional cost of supply model. General customer classes that are connected to the lower voltage parts of the network receive average network prices consisting of combinations of energy and demand price structures.

For further information please refer to the Network Strategy Statement.

References:

Section 2.2 –
Country
Energy's
Network
Strategy
Statement

6. Provided that economic costs are covered, prices should be responsive to the requirements and circumstances of users in order to:

(a) discourage uneconomic bypass

(b) allow negotiation to better reflect the economic value of specific services, including services associated with embedded generation and other options

Network prices reflect the standard level of electricity supply in terms of quality, reliability and security available to the customer at their point of connection to the network. For major subtransmission customers, the network prices have been calculated on a cost reflective basis. However customers connected further into the distribution system will see a greater degree of averaging in their network prices.

Opportunities may arise for the customer to negotiate a network price, which is different from the published prices, where special circumstances exist. IPART has indicated that customers may have access to negotiated pricing arrangements under regulatory guidelines.

Where customers believe that they have been, or will be, disadvantaged by the methodology applied by Country Energy to determine use of system charges and that the above criteria is met, a review of the customer's circumstances will be undertaken. The conditions negotiated would normally be embodied in a network connection agreement.

7. When allocating TUOS charges to distribution network users distributors should, to the extent possible, preserve the pricing signals present in the structure of TUOS charges. (Information on allocated TUOS charges should be available to users on request, to the extent possible)

Transmission charges do not have a form that readily translates into network price structures. The pass-through of transmission charges and the structure of network prices have been aligned where possible. However, direct mapping to network prices for standard customer classes has not been possible due to the large, fixed transmission charges that cannot be directly included in network price structures, which typically have a small fixed charge.

Country Energy has separated network prices into DUOS and transmission cost recovery prices using the principles proposed by the distributors in the Joint Allocation Methodology. The principles from that methodology have been generally applied by Country Energy.

The allocation of transmission related costs to prices is described in further detail in the Network Strategy Statement.

References:

Sections 2.4,
2.6.5, 2.9.6,
2.9.8 –
Country
Energy's
Network
Strategy
Statement

8. Information on customer class price levels and structures, service standards, underlying costs, price derivation methods and rationale and medium term price and service strategies should be publicly disclosed

Public disclosure should allow current and potential users to understand the basis for prices and to take account of prices and service standards in their consumption, investment and location decisions. In addition, disclosure should also allow interested parties to better assess the range of opportunities for meeting user requirements, including through services associated with embedded generation, demand management and other options that may reduce users' costs and lead to more efficient outcomes. The Network Strategy Statement provides detailed information in this regard.	References: Sections 2, 3, 4 – Country Energy's Network Strategy Statement
9. Underlying service classifications, cost data, cost allocations and other elements that contribute to pricing decisions should be periodically reviewed and updated where relevant to reflect industry developments and changes in user requirements and preferences, methods of service provision and costs	
Country Energy will endeavour to periodically review and update its network prices and service standards for prescribed distribution services to reflect this requirement. Network prices and service standards are generally reviewed on an annual basis as part of the business planning process. For further information please refer to the Network Strategy Statement and this Annual Prices Report.	References: Section 2.11.6 – Country Energy's Network Strategy Statement
10. Where distributor price strategies lead to proposed price movements or price restructuring that may be expected to impose significant adjustment costs on users, transitional price options, a phased approach or other measures should be offered to assist in the management of adjustment costs.	
Country Energy will consider options for moving customers to transitional network pricing options during the regulatory control period to cushion the financial impact resulting from significant network price adjustments. Country Energy does not expect that the application of its medium-term pricing strategy will impose significant adjustment costs on customers, as prices will move in accordance with the limits on price movements as regulated by IPART. Transitional network prices are not available in 2005/06. For further information please refer to the Network Strategy Statement.	References: Section 2.11.7 – Country Energy's Network Strategy Statement

7 Expected levels of service and projected capital expenditure projects

The NSW Electricity Distribution Pricing 2004/05 to 2008/09 Determination requires that an annual prices document submitted by a DNSP under clause 11 outlines the expected levels of service and projected capital expenditure projects that will occur for the coming year.

7.1 Capital expenditure projects

Capital expenditure is invested efficiently by Country Energy to deliver the greatest benefit to all stakeholders. Our medium-term capital investment program is designed to meet targeted programs and to continue to deliver the current levels of service to customers. Specifically the program takes into account the following factors:

- The needs of our customers and all other stakeholders
- Current and projected levels of service standards, including long-term network reliability across the network and the specific targeting of those parts of the network that are delivering the lowest service levels

- Independently assessed forecasts of demand growth
- Condition and age of assets
- Regulatory, environmental and safety compliance requirements
- Non system investment requirements
- Efficient alternatives to capital expenditure including efficient operating and maintenance expenditure, demand management, and new generation.

As part of the recent electricity distribution pricing review, IPART undertook a review of the forecast capital expenditure and engaged engineering consultants to undertake a detailed analysis of capital expenditure requirements. The consultants endorsed the levels of capital expenditure proposed by Country Energy and concluded that the investment program is efficient and justified. The consultants also independently reviewed capital expenditures incurred over the previous regulatory period and concluded that capital expenditure incurred during the period was prudent. IPART has adopted the capital expenditure allowances for Country Energy that align with the recommendations of their consultants.

The following section outlines the major capital expenditure categories and provides a brief description of major capital expenditure projects planned for the 2005/06 financial year necessary to meet:

- Growth in demand and system augmentation requirements to maintain system performance and system security
- Asset replacement and renewal requirements to maintain system performance, safety and integrity of the network and to meet customer requirements for network quality and reliability
- Non-system investment requirements to support the operations of the distribution business.

7.1.1 Growth in demand and system augmentation

Growth-related expenditure relates to investments that are required to ensure that the network has adequate capacity to meet increasing demand.

Country Energy's distribution network is operating at or near its maximum capacity in some areas. The strong growth areas include the major regional centres and the northern and southern coastal strips, the latter areas representing some of the fastest growing areas in the state. This reflects continued residential and commercial development, with expansion forecast to continue into the future. Over the medium term, Country Energy will need to continue to expand and reinforce the network in these areas to meet the high growth in demand. In regions west of the ranges, the general rate of growth in residential development is moderate.

A selection of specific growth-related projects for 2005/06 are outlined below-

➤ Terranora network area

The existing Terranora 33,000-volt network in the Far North Coast region of NSW is reaching full design capacity. The most efficient approach to relieve emerging network constraints in this area is to increase the subtransmission voltage from a current rating of 33,000 volts to 66,000 volts. A number of sections of this subtransmission network are constructed to allow operation at 66,000 volts. Conversion to full 66,000 volt operation forms part of the strategic network development plan for the Terranora supply area.

Included in this plan is the conversion of the Cudgen Zone Substation to 66,000-volt operation, involving the installation of two new 66/11,000 volt transformers. The project also includes the establishment of a 33,000 volt supply from the substation, involving the installation of two 66/33,000 volt transformers and the reconstruction of the 11,000-volt switchboard and control building. The cost of this project is estimated to be \$4.87 million.

Another related project involves the uprating of the existing 33,000 volt switchgear at Terranora zone substation to 66,000 volt. The project cost is estimated to be \$1.6 million.

This work commenced in 2002/03 and is scheduled to be completed in 2005/06.

➤ **Mullumbimby network area**

Strong load growth is currently being experienced on the Far North Coast. Existing load within the Ocean Shores/Mullumbimby/Ewingsdale area is currently supplied primarily from the Mullumbimby Substation, with redundancy provided by the 11kV network from Ewingsdale.

The expected load growth in the area will limit redundancy capability via the 11kV network. To increase the reliability of supply to the Mullumbimby 11kV and the Lismore 66kV ring, an additional transformer is required to be purchased for the Mullumbimby substation.

The project cost is estimated to be \$3.4 million.

➤ **Wagga Wagga network area**

Wagga Wagga is currently supplied by 13 zone substations operating at 66,000 volts. The loss of a single subtransmission feeder can result in the loss of 50 per cent of the supply to four major zone substations in the area. The creation of a 66,000-volt ring system would eliminate the potential risk of loss of supply resulting from the failure of a single subtransmission network element. Country Energy's subtransmission planning guidelines requires zone substations loaded to greater than 15 MVA to be upgraded to meet an N-1 reliability criterion. Augmentation for thermal rating will be required at the Hammond Avenue, Ashmont, Bomen, and Bourkelands zone substations.

The project involves the augmentation of the Hammond Avenue 66/11,000 volt substation to allow ring feed conversion and northern supply point connection. The project cost is estimated to be \$2.2 million.

Work commenced at Hammond Avenue in 2004/05 and is expected to be completed during 2005/06.

➤ **Martins Creek/Gresford network area**

Martins Creek 33/11kV and Gresford 33/11kV zone substations are currently supplied by a single 33kV supply from EnergyAustralia's East Maitland substation. Incoming voltage is weak at Martins Creek and a number of reliability issues have led to extended outages for customers supplied by both Martins Creek and Gresford Substations. A new 33kV line between Woerden and Martins Ck (12km) will be constructed and an existing 66kV line between Woerden and Dungog will be utilised at 33kV to provide an alternate 33kV supply to Martins Creek. The project cost is estimated to be \$2.0 million.

➤ **Bulahdelah network area**

The existing supply point at Stroud is the main injection point into the Bulahdelah ring system which supplies Bulahdelah, Pacific Palms and Booral and provides alternative supply to Tea Gardens. This supply is via two 33kV feeders which are

required to supply loads up to 40 – 60 kilometres from Stroud. The levels of loading on this ring system are such that the incoming voltage to Bulahdelah is poor during system normal conditions. During outage conditions, the incoming voltage to Bulahdelah is well below planning standards. Additionally, during outage conditions, thermal constraints exist on the line remaining in service. It is proposed to commission a new 132/33kV bulk supply point at Bulahdelah to avoid the augmentation of both Bulahdelah feeders and the Stroud supply point. This would also provide an option for normal supply to Tea Gardens. It is planned to purchase a site at Bulahdelah during 2005/06 at an estimated cost of \$150,000. The zone substation is expected to be constructed during 2006/07.

➤ **Wentworth network area**

The Wentworth region is serviced by cross-border supplies from Victoria. One 66kV feeder supplies the Dareton 66/22kV zone substation. This substation supplies four 22kV feeders, which service the Wentworth/Buronga/Gol Gol area. Some of these 22kV feeders go on to supply long 19.1kV SWER lines. There is a limited back-up 22kV cross-border supply at Gol Gol.

The Dareton Zone Substation is approximately midway between the Wentworth and Buronga load centres, with significant growth occurring in both areas. A new mineral sands mining industry is to commence near Pooncarie (approximately 110km north of Wentworth) during 2005/06. Major works include:

- Merbein – Dareton 66 kV supply power factor correction (\$1.5m)
- Dareton – Ellerslie 66 kV supply (\$2.7m)
- Wentworth 22 kV feeder augmentation (\$0.7m)
- Establish new Buronga/Gol Gol depot (\$0.6m)

➤ **Broken Hill network area**

The northern part of the Far West region is supplied from TransGrid's Broken Hill 220/22kV zone substation. Six 22kV feeders supply the local area, and two of these (one plus backup) also supply the Country Energy Pinnacles Place 22/66kV Zone Substation. From this zone substation a single radial 66kV sub-transmission line supplies six more zone substations, which in turn supply 33kV, 19.1kV SWER, 22kV and 3.3kV feeders. These feeders supply the remainder of the northern region.

They are almost exclusively radial and have main line route lengths of up to 300 kilometres. One of the 33kV feeders supplies a seventh zone substation, which in turn supplies the Wilcannia township at 6.6kV.

Major works include:

- Insulate earthed air break switches (\$0.3m)
- Multiple Earthed Neutral (MEN) conversion (\$0.3m)
- 66 kV feeder Kinalung – Sunset Strip upgrade (\$1.2m)
- Zone substation protection and fencing upgrades (\$0.95m)

7.1.2 Asset renewal and replacement

A significant component of the overall network-related capital program is the replacement and renewal of aged assets. Network assets have a finite life. Replacement and renewal investment ensures the timely replacement or refurbishment of network assets that have become unserviceable, that frequently fail in service, have significantly deteriorated to an unsafe or risky condition or where the

present-value cost of maintaining the asset exceeds the cost of replacement. Asset replacement expenditure increases gradually over time due to the ageing of the network. This trend is already evident and will continue into the new regulatory period. Where the condition of the assets has deteriorated and ongoing maintenance is not effective, system assets must be replaced.

Replacement is also driven by environmental regulatory and industry safety compliance considerations. General distribution related reliability capital expenditure to maintain existing service levels are also included in this category. More than 40 per cent of Country Energy's network assets are more than 40 years old and are rapidly ageing. In the areas of the network west of the ranges, the network assets are generally older and the condition of some assets are increasingly requiring replacement in order to maintain the performance level of the network.

7.1.3 Non-system requirements

This category includes capital items that will be required to support the electricity distribution business including:

- Information systems development and computer hardware, software and upgrades, licences, plant and equipment relating to IT assets, and related expenditure required to operate the business
- Motor vehicles and plant
- Communication systems such as trunk mobile radio, faxes, mobile phones
- A number of other miscellaneous expenditures such as general office equipment, furniture and whitegoods, small plant and test equipment, and building and property works.

Expenditure related to this category will ensure the continued efficient management and support of the electricity distribution business.

7.2 Expected levels of service for the coming year

Country Energy is proud of the service standards achieved within a difficult and unique operating environment.

During the recent electricity distribution pricing review, IPART accepted Country Energy's reliability performance objectives for the five-year period commencing July 2004. The objectives are based on actions being undertaken by Country Energy to improve the availability and robustness of reliability data through the implementation of an enhanced reliability reporting system and the implementation of the expenditure program.

The SAIDI reliability performance objectives for 2005/06 for different components of the network are tabled below. SAIDI represents the total number of minutes on average that each customer installation connection to the distribution system (or to those components) is without a supply of electricity.

Figure 7 SAIDI Reliability Performance Objectives for 2005/06

Component of distribution system	2005/06 SAIDI objective (minutes)
Whole distribution system	361
CBD feeders	N/A
Urban feeders	129
Rural short feeders	371
Rural long feeders	757

8 Expected consumption

Country Energy's regions have features that differentiate them from the rest of the New South Wales economy in general. There are also significant differences across the various regions that make up Country Energy.

Energy sales in the residential and commercial sectors is dependent on the wellbeing of the rural economies and the fortunes of rural-based industries such as agriculture, food processing, manufacturing, mining and resource processing.

Country Energy's service area is also characterised by the dominance of large point loads in specific industries, most of which are exposed to international trade and exchange rate fluctuations. The isolation of many of their locations intensifies the dependence of the local economy on these industries. Energy consumption and growth in the residential and commercial customer sectors therefore tends to reflect the movement in these industries. There is also an under representation of service industries in regional and rural areas, which is more typical of metropolitan areas. In addition, Country Energy's vast distribution area is exposed to environmental and climatic factors, which can impact on energy sales. Covering 95 per cent of New South Wales' land mass, our network spans mountains, plains and deserts, with climates ranging from snow to extreme heat. In some instances, these factors can be a key factor in the volatility of sales.

The mix of economic activities within our distribution area has resulted in less diversification in the area's economy, when compared to the New South Wales economy generally and those of the other major distributors.

In consideration of these issues, Country Energy engaged the National Institute of Economic and Industry Research (NIEIR), an independent economic forecaster, to provide electricity growth forecasts. The NIEIR study included economic, energy, customer connections and peak demand forecasts for Country Energy for three alternative economic scenarios. Further analysis of the NIEIR projections was carried out by Country Energy. This information was used in our submissions to IPART during the recent electricity distribution pricing review and for other business planning purposes.

The forecasting methodologies employed by NIEIR are theoretically sound and proven and have been employed throughout the electricity industry for some time. The forecasts differentiate between expected growth across different customer classes.

8.1 Socio-economic forecasts

The strongest area of economic growth is the northern region of Country Energy. The weakest areas of economic growth are located west of the ranges. Country Energy's distribution area's economic growth is supported by steady growth in the mining and agricultural areas.

Total population within our distribution area is expected to increase by an average 0.5 per cent per annum to 2009. Growth is dominated in the north coast region at 0.7 per cent per annum. The New South Wales average is 0.9 per cent per annum.

The total number of households within our distribution area is forecast to grow by an average of 1.1 per cent to 2009. This falls under the New South Wales average by 0.4 per cent. Growth is also lead by the north coast region at 1.4 per cent per annum.

Projections of total gross regional product (GRP) for the Country Energy distribution area is expected to rise by an average of 2.2 per cent. This is below the New South Wales average by some 0.8 per cent however, the north coast region of Country Energy is a notable exception, at 3.0 per cent per annum.

8.2 Expected growth in electricity sales by customer class

Under the medium-growth scenario, total electricity sales growth is expected to average 1.7 per cent to 2009. Total energy consumption is projected to increase to 11,043 GWh in 2005/06.

Residential sales growth over recent years has been supported by the recovery in dwelling construction as well as higher penetrations of air conditioning equipment (following two hot summers over recent years). Residential sales growth in the Country Energy distribution area is expected to average 1.6 per cent per annum to 2009 under the medium-growth scenario. Total residential energy consumption is expected to be 4,875 GWh in 2005/06.

Business sales growth largely mirrors the profile for Country Energy's gross state product growth. Average sales growth in the business class is expected to be 1.7 per cent to 2009 under the medium growth scenario. Total business energy consumption is expected to be 6,168 GWh in 2005/06.

8.3 Expected growth in electricity customers

The pattern of growth in residential customers reflects dwelling construction projection. The projections of business customers takes into account recorded historical growth rates, expected economic activity, as well as cross checks against other country areas.

Under the medium-growth scenario, total customer numbers are expected to increase at an average rate of 1.4 per cent per annum to 2009. Residential customer numbers are expected to grow at 1.6 per cent per annum and business customer numbers are expected to grow at 0.8 per cent per annum to 2009.

8.4 Expected growth in maximum demand

Peak demand forecasts are a key element in determining Country Energy's capital expenditure requirements in specific locations, in order to meet the expected growth in demand.

Growth in maximum demand from residential customers is primarily due to the connection of new customers to the network and the increased uptake of air

conditioners and other household electrical equipment. The prime drivers of demand by business customers include the number of new connections and general economic conditions. Another prime driver of demand is weather, particularly temperature, which is used as an input to system demand forecasts.

Maximum summer and winter demand forecasts have been developed for the total Country Energy distribution area for alternative probabilities of exceedence 10 per cent (a one in ten year chance of occurring), a 50 per cent probability (one in two year chance of occurring) and a 90 per cent probability (nine in ten year chance of occurring). Within these two sets of forecasts are forecasts for summer and winter.

For the 50th percentile maximum demand under the medium economic growth scenario:

- The winter maximum demand rises at an average annual rate of 2.8 per cent per annum
- The summer maximum demand rises at an average annual rate of 3.2 per cent per annum

The maximum winter demand for the whole network is forecast to grow to 2,154 MW and the maximum summer demand is forecast to grow to 1,798 MW in 2005/06. In particular, the northern coastal region is expecting substantial demand growth, mainly due to residential developments.